ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2019 - 2020)

NEET(UG)
MINOR
01-12-2019

PRE-MEDICAL : ENTHUSIAST COURSE (PHASE : MEQ)

*This Booklet contains **20** pages.*

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is ***not*** permissible on the Answer Sheet.

Name of the Candidate (in Capitals)_____

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Your Target is to secure Good Rank in Pre-Medical 2020

TOPIC : Kinematics, Projectile Motion, Relative Motion, Gravity,SHM, Wave Motion, NLM, Friction & WPE

1. A particle moves along a straight line. Its position at any instant is given by $x = 32t - \frac{8t^3}{3}$ where x is in metres and t in seconds. Find the acceleration of the particle at the instant when particle is at rest :-

(1) – 16 ms^{-2} (2) – 32 ms^{-2} (3) 32 ms^{-2} (4) 16 ms^{-2}

2. A ball is released from the top of a tower of height h meter. It takes T second to reach the ground. What is the position of the ball in T/3 second ?

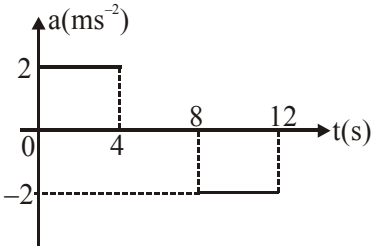
(1) $\frac{h}{9}$ meter from the ground

(2) (7h/9) metre from the ground

(3) (8h/9) meter from the ground

(4) (17h/18) meter from the ground

3. A lift starts from rest. Its acceleration is plotted against time. When it comes to rest its height above its starting point is :-



(1) 20 m (2) 64 m (3) 32 m (4) 36 m

4. A particle moving in a straight line has velocity-displacement equation as $V = 5\sqrt{1+s}$. Here V is in ms^{-1} and s in metres. Select the correct alternative.

(1) Particle is initially at rest.

(2) Initially velocity of the particle is 5m/s and the particle has a constant acceleration of 12.5 ms^{-2}

(3) Particle moves with a uniform velocity

(4) None of the above

5. If T_1 and T_2 are the times of flight for two complementary angles, then the range of projectile R is given by :-

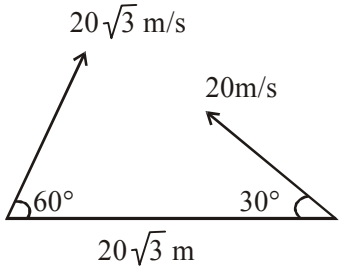
(1) $R = 4gT_1T_2$ (2) $R = 2gT_1T_2$

(3) $R = \frac{1}{4}gT_1T_2$ (4) $R = \frac{1}{2}gT_1T_2$

6. Two stones are projected with the same speed but making different angles with the horizontal. Their horizontal ranges are equal. The angle of projection of one is $\frac{\pi}{3}$ and the maximum height reached by it is 102 m. Then the maximum height reached by the other in metres is :-

(1) 76 (2) 84 (3) 56 (4) 34

7. In the figure shown, the two projectiles are fired simultaneously. The minimum distance between them during their flight is :-



(1) 20 m (2) $10\sqrt{3}$ m

(3) 10 m (4) None of the above

8. In damped oscillations, the amplitude of oscillations is reduced to one-third of its initial value a_0 at the end of 100 oscillations. When the oscillator completes 200 oscillations, its amplitude must be :-

(1) $a_0/2$ (2) $a_0/6$ (3) $a_0/12$ (4) $a_0/9$

9. A body initially at mean position oscillates simple harmonically with a period of 2s. After what time will its kinetic energy be 75% of the total energy:-

(1) 1/12 sec (2) 1/6 sec

(3) 1/4 sec (4) 1/3 sec

10. A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released, so that the mass executes SHM of time period T. If the mass is increased by m, the time period becomes 5T/3. The ratio of m/M is:-

(1) $\frac{5}{3}$ (2) $\frac{3}{5}$ (3) $\frac{16}{9}$ (4) $\frac{25}{9}$

11. A locomotive engine approaches a railway station and whistles at a frequency of 400 Hz. A stationary observer on the platform observes a change of 40 Hz as the engine passes across him. If the velocity of sound is 340 m/s, the speed of the engine is :-

(1) 34 m/s (2) 40 m/s (3) 17 m/s (4) 20 m/s

12. The angle made by the string as a simple pendulum with the vertical depends upon time as

$\theta = \frac{\pi}{90}\sin \pi t$ Find length of the pendulum :-

(1) 0.5 m (2) 1 m (3) 1.5 m (4) 1.2 m

13. If the equation of transverse wave is

$Y = 5\sin 2\left(\pi\frac{t}{0.04} - \frac{x}{40}\right)$

where distance is in cm and time in second, then the wavelength of wave will be

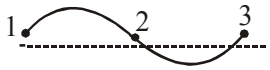
(1) 20 cm (2) 40 cm

(3) 60 cm (4) None of these

14. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of open pipe. The fundamental frequency of open pipe is :-

(1) 200 Hz (2) 300 Hz (3) 240 Hz (4) 480 Hz

15. The diagram shows an instantaneous position of a string as a transverse progressive wave travels along it from left to right. Which one of the following correctly shows the direction of velocity of points 1, 2 and 3 on the string?



	1	2	3
(1)	↑	↓	↓
(2)	↑	↓	↑
(3)	↓	↓	↓
(4)	↓	↑	↓

16. Sound velocity is maximum in :-

(1) H_2 (2) N_2

(3) He (4) O_2

17. Two waves represented by;

$y_1 = 10\sin 2000\pi t$, $y_2 = 20\sin\left(2000\pi t + \frac{\pi}{2}\right)$ are superimposed at any point at a particular instant. The amplitude of the resultant wave is :

(1) 200 (2) 30 (3) $10\sqrt{5}$ (4) $10\sqrt{3}$

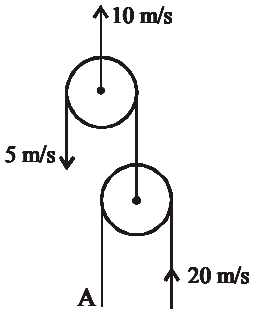
18. The two waves having intensities in the ratio 1 : 9 produce interference. The ratio of the maximum to the minimum intensities is equal to :

(1) 10 : 8 (2) 9 : 1 (3) 4 : 1 (4) 2 : 1

19. An accurate and reliable audio oscillator is used to standardize a tuning fork. When the oscillator reading is 514, two beats are heard per second. When the oscillator reading is 510, the beat frequency is 6 Hz. The frequency of the tuning fork is :-

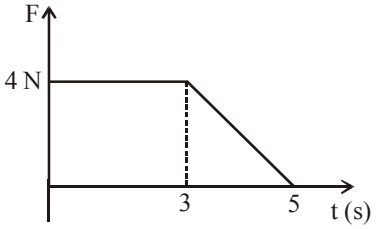
(1) 506 (2) 510 (3) 516 (4) 512

20. Velocity of A will be :-



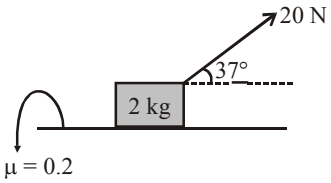
(1) 35 m/s (2) 25 m/s (3) 30 m/s (4) 20 m/s

21. A force is applied on a block of 2 kg , force vs time graph is given, if initial velocity of block is 5m/s then find velocity of block at t = 5 sec



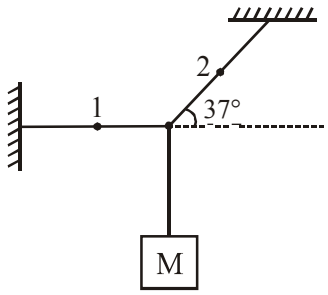
(1) 8 m/s (2) 13 m/s (3) 16 m/s (4) 17 m/s

22. Initially block was in rest then velocity of block at t = 5 sec will be :-



(1) 7.2 m/s (2) 35 m/s (3) 36 m/s (4) 7 m/s

23. Find out ratio of tension in string (1) and string (2) T_1/T_2



(1) $\frac{3}{5}$ (2) 1 (3) $\frac{5}{3}$ (4) $\frac{4}{5}$

24. Ball of mass 2 kg moving with speed of 4 m/s strikes the wall normally and rebound with speed of 2 m/s , normally to the wall find magnitude of impulse given by wall to the ball.

(1) 4 N.s (2) 12 N.s (3) 8 N.s (4) 16 N.s

25. Select incorrect statement from the following :-

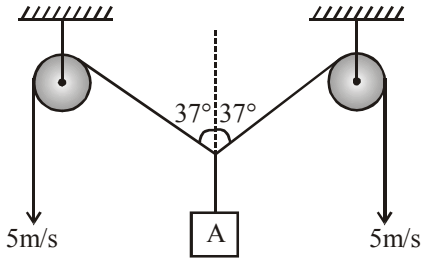
(1) Tension and normal may be impulsive force.

(2) Gravitational and spring force always non-impulsive.

(3) Only impulsive forces can provide impulse.

(4) Non-impulsive forces can also provide impulse.

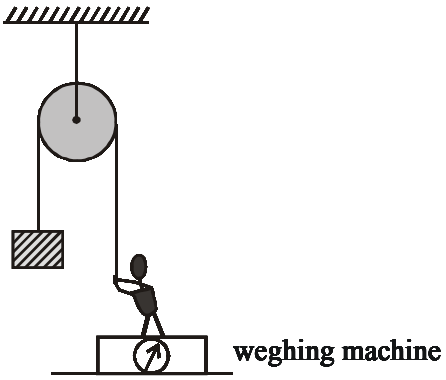
26. In give arrangements speed of block A will be :-



(1) 4 m/s (2) 8 m/s

(3) $\frac{50}{4}$ m/s (4) $\frac{25}{4}$ m/s

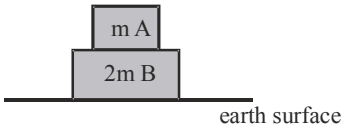
27. In given arrangement mass of block is 5 kg, mass of man is 50 kg man is pulling block with acceleration of 6 m/s^2 then reading of w.m. will be :-



(1) 80 N (2) 580 N

(3) 420 N (4) 450 N

28. Block A and B are placed on earth surface as shown in figure select correct alternative.



(1) Net force acting by earth on block A is zero.

(2) Net force acting by earth on Block B is zero.

(3) Net force acting by block B on earth is mg downward.

(4) Net force acting by block B on block A is mg downward.

29. Power supplied to a particle of mass 2kg varies with time as $P = \frac{3t^2}{2}W$. Here t is in second. If the velocity of particle at t = 0 is v = 0, the velocity of particle at time t = 2s will be :-

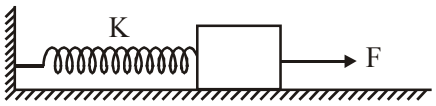
(1) 1 m/s

(2) 4 m/s

(3) 2 m/s

(4) $2\sqrt{2}$ m/s

30. A block attached to a spring, pulled by a constant horizontal force F, is kept on a smooth surface.



Initially, the spring is in the natural length state. Then the maximum positive work that the applied force F can do :-

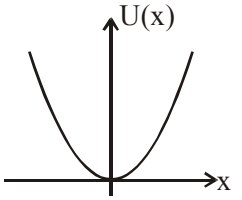
(1) $\frac{F^2}{K}$ (2) $\frac{2F^2}{K}$

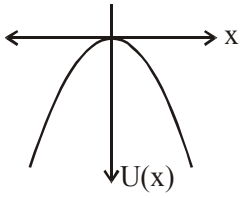
(3) ∞ (4) $\frac{F^2}{2K}$

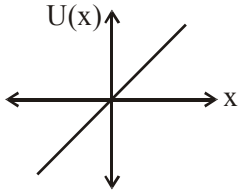
31. A wind powered generator converts wind energy into electrical energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to :-

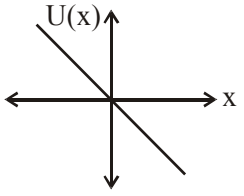
(1) v (2) v^2 (3) v^3 (4) v^4

32. A particle is acted by a force F = kx, where k is a positive constant. Its potential energy at x = 0 is zero. Which curve correctly represente the variation of potential energy of the block with respect to x ?

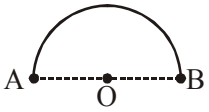
(1) 

(2) 

(3) 

(4) 

33. A uniform thin rod of mass m and length l is bent into a semicircle. Gravitational field intensity at the centre O is :-



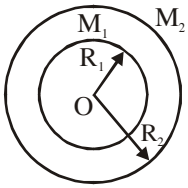
(1) $\frac{2\pi Gm}{l^2}$ along AOB

(2) $\frac{\pi Gm}{l^2}$ perpendicular to AOB

(3) $\frac{2\pi Gm}{l^2}$ perpendicular to AOB

(4) $\frac{\pi Gm}{l^2}$ along AOB

34. Figure shows two concentric spherical shells of masses M_1 and M_2 and of radii R_1 and R_2, respectively. Gravitational potential at a point at a distance x from centre such that $R_1 < x < R_2$ will be:-



(1) $\frac{-GM_2}{R_2}$

(2) $-G\left(\frac{M_1}{R_1}+\frac{M_2}{x}\right)$

(3) $-G\left(\frac{M_1}{x}+\frac{M_2}{R_1}\right)$

(4) $-G\left(\frac{M_1}{x}+\frac{M_2}{R_2}\right)$

35. A body weighing 20 kg on the surface of the earth is taken to a place 6000 km below the earth's surface. Assuming the earth's radius to be 6000 km, the weight of the body at that depth is :-

(1) 40 kg (2) 5 kg

(3) infinite (4) Zero

36. A body of mass m is placed on earth surface which is taken from earth surface to a height of h = 3R, then change in gravitational potential energy is :-

(1) $\frac{mgR}{4}$ (2) $\frac{2mgR}{3}$

(3) $\frac{3mgR}{4}$ (4) $\frac{mgR}{2}$

37. If an artificial satellite is moving in a circular orbit around the earth with a speed equal to half the magnitude of escape velocity from the surface of earth, height of the satellite above the surface of earth is :-

(1) 2R (2) $\frac{R}{2}$

(3) R (4) $\frac{R}{4}$

38. A particle is thrown with escape velocity v_e = 11.2 km/s from the surface of earth. Its velocity at height 3R = 19200 km is :-

(1) 9.25 km/s (2) 5.6 km/s (3) 11.2 km/s (4) 4.3 km/s

39. Halley's comet has a period of 76 years and in 1986, had a distance of closest approach to the sun equal to 8.9×10^{10} m. What is the comet's farthest distance from the sun if the mass of sun is 2×10^{30} kg and $G = 6.67 \times 10^{-11}$ MKS units ?

(1) 2.7×10^{12} m (2) 5.3×10^{12} m

(3) 8.9×10^{10} m (4) 6.8×10^{10} m

40. A magnet of magnetic moment $50\hat{i}\ A-m^2$ is placed along the x-axis in a magnetic field $\vec{B}=(0.5\hat{i}+3.0\hat{j})T$. The torque acting on the magnet is :

(1) $175\ \hat{k}\ N-m$ (2) $150\ \hat{k}\ N-m$

(3) $75\hat{k}\ N-m$ (4) $25\sqrt{37}\hat{k}\ N-m$

41. At magnetic poles of earth, angle of dip is :

(1) Zero (2) 45° (3) 90° (4) 180°

42. The radius of the coil of a Tangent galvanometer, which has 10 turns, is 0.1m. The current required to produce a deflection of 60° is ($B_H = 4\times10^{-5}T$) :

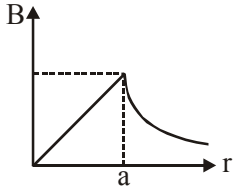
(1) 3A (2) 1.1A (3) 2.1A (4) 1.5A

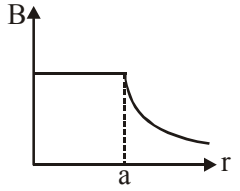
43. A proton carrying 1MeV kinetic energy is moving in a circular path of radius R in uniform magnetic field. What should be the energy of an α-particle to describe a circle of the same radius in the same field ?

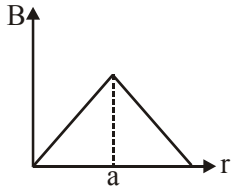
(1) 2 MeV (2) 1 MeV

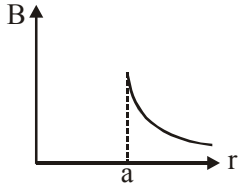
(3) 0.5 MeV (4) 4 MeV

44. The magnetic field due to a straight conductor of uniform cross section of radius a and carrying a steady current is represented by :

(1)


(2)


(3)


(4)


45. Two parallel beams of positrons moving in the same direction will :

(1) attract each other

(2) repel each other

(3) not interact with each other

(4) be deflected normal to the plane containing the two beams

TOPIC : Some Basic Concepts of Chemistry, Solid State, Surface Chemistry, Chemical Equilibrium and Ionic Equilibrium

46. Which of the following is not a characteristic of chemisorption?
(1) Adsorption is irreversible.
(2) Adsorption decreases with increase in temperature.
(3) Adsorption is specific.
(4) Adsorption increases with increase of surface area.

47. On addition of one ml solution of 10% NaCl to 10 ml red gold sol in the presence of 0.0025 g of starch, the coagulation is just prevented. The gold number of starch is :-
(1) 25 (2) 2.5 (3) 0.25 (4) 0.025

48. Which of the following is most effective in causing the coagulation of ferric hydroxide sol ?
(1) KCl (2) KNO_3
(3) K_2SO_4 (4) $K_3[Fe(CN)_6]$

49. An emulsion is a colloidal solution consisting of:-
(1) two solids
(2) two liquids
(3) two gases
(4) one solid and one liquid

50. Which of the following substance give a positively charged sol ?
(1) Gold
(2) Arsenious sulphide
(3) Starch
(4) Ferric hydroxide

51. Surface tension of lyophilic sol is :-
(1) lower than that of dispersion medium
(2) more than that of dispersion medium
(3) equal than that of dispersion medium
(4) any of these, depending on the dispersion medium

52. When dilute aqueous solution of $AgNO_3$ (excess) is added to KI solution, positively charged sol particles of AgI are formed due to adsorption of ion ?
(1) K^+ (2) Ag^+ (3) I^- (4) NO_3^-

53. The simplest formula of a solid having CCP arrangement for 'A' atoms in which alternate face-centres are occupied by 'B' atoms and alternate edge centres are occupied by 'C' atoms, is :-
(1) ABC (2) A_3BC (3) A_2BC (4) A_4B_2C

54. The distance between two nearest neighbours in BCC lattice of axial length, l is :-
(1) l (2) $\frac{\sqrt{3}}{2}l$ (3) $\frac{\sqrt{2}}{2}l$ (4) $\frac{1}{2}l$

55. The number of tetrahedral and octahedral voids in hexagonal primitive unit cell is :-
(1) 8, 4 (2) 4, 8 (3) 12, 6 (4) 6, 12

56. The radius of Ag^+ ion is 126 pm while of I^- ion is 216 pm. The co-ordination number of Ag in AgI is :-
(1) 2 (2) 4 (3) 6 (4) 8

57. A-definite amount of solid NH_4HS is placed in a flask already containing ammonia gas at a certain temperature and 0.50 atm pressure. NH_4HS decomposes to give NH_3 and H_2S and at equilibrium total pressure in flask is 0.84 atm. The equilibrium constant (K_P) for the reaction is :-
(1) 0.30 (2) 0.18 (3) 0.17 (4) 0.11

58. On applying pressure to the equilibrium,

$$\text{Ice} \rightleftharpoons \text{Water}$$

Which phenomenon will happen ?
(1) More ice will be formed
(2) More water will be formed
(3) Equilibrium will not be disturbed
(4) Water will evaporate

59. Amount of PCl_5 (in mole) need to be added to one litre vessel at 250°C in order to obtain a concentration of 0.1 mole of Cl_2 for the given change is :-

$$PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)} \; ; \; K_c = 0.0414$$

(1) 0.3415 (2) 0.0341
(3) 3.415 (4) 0.3145

60. The conjugate acid of HPO_4^{2-} is :-

(1) $H_2PO_4^-$ (2) PO_4^{3-}

(3) H_3PO_4 (4) H_3PO_3

61. Species acting as both bronsted acid and base is :-

(1) HSO_4^- (2) Na_2CO_3

(3) $\overset{\oplus}{N}H_4$ (4) OH^-

62. Given $HF + H_2O \xrightarrow{K_a} H_3O^+ + F^-$

$F^- + H_2O \xrightarrow{K_b} HF + OH^-$

Which relation is correct ?

(1) $K_b = K_w$ (2) $K_b = 1/K_w$

(3) $K_a \times K_b = K_w$ (4) $K_a/K_b = K_w$

63. How many litres of water must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2 ?

(1) 0.9 L (2) 2.0 L (3) 9.0 L (4) 0.1 L

64. The total number of different kind of buffers obtained during the titration of H_3PO_4 with NaOH are :-

(1) 3 (2) 1 (3) 2 (4) Zero

65. 10 mL of a solution contains 0.1 M NH_4Cl + 0.01 M NH_4OH. Which addition would not change the pH of the solution :-

(1) Adding 1 mL water

(2) Adding 5 mL of 0.1 M NH_4Cl

(3) Adding 5 mL of 0.1 M NH_4OH

(4) Adding 10 mL of 0.1 M NH_4Cl

66. The H^+ ion concentration in 0.001 M acetic acid is 1.34×10^{-4} g ion/litre. The H^+ ion concentration when 0.164 g of CH_3COONa is added to a litre of 0.001 M CH_3COOH will be :-

(1) 9×10^{-6} (2) 18×10^{-6}

(3) 4.5×10^{-6} (4) 5×10^{-6}

67. The precipitate of CaF_2 ($K_{sp} = 1.7 \times 10^{-10}$) is obtained when equal volumes of the following are mixed :-

(1) 10^{-4} M Ca^{2+} + 10^{-4} M F^-

(2) 10^{-2} M Ca^{2+} + 10^{-3} M F^-

(3) 10^{-5} M Ca^{2+} + 10^{-3} M F^-

(4) 10^{-3} M Ca^{2+} + 10^{-5} M F^-

68. A student wants to prepare a saturated solution of Ag^+ ion. He has got three samples AgCl ($K_{sp} = 10^{-10}$), AgBr ($K_{sp} = 10^{-13}$) and Ag_2CrO_4 ($K_{sp} = 10^{-12}$). Which of the above compound will be used by him using minimum weight to prepare the saturated solution ?

(1) AgCl

(2) AgBr

(3) Ag_2CrO_4

(4) All of these

69. The solubility products of $Al(OH)_3$ and $Zn(OH)_2$ are 8.5×10^{-23} and 1.8×10^{-14} respectively. If NH_4OH is added to a solution containing Al^{3+} and Zn^{2+} ions, then substance precipitated first is :-

(1) Al $(OH)_3$

(2) $Zn(OH)_2$

(3) Both together

(4) None of these

70. The dissociation constant of HCN is 1.3×10^{-9}. The value of hydrolysis constant of KCN will be :-

(1) 1.3×10^{-9} (2) 10^{-14}

(3) 7.7×10^{-5} (4) 0.77×10^{-5}

71. 0.1 M acetic acid solution is titrated against 0.1 M NaOH solution. What would be the difference in pH between 1/4 and 3/4 stages of neutralisation of acid ?

(1) 2 log 3/4 (2) 2 log 1/4

(3) log 1/3 (4) 2 log 3

72. The volume in litres of CO_2 liberated at NTP when 10g of 80% pure lime stone is heated completely is :-

(1) 22.4 L (2) 2.24 L

(3) 20.16 L (4) 1.8 L

73. At NTP, 5.6 L of a gas weight 8g. The vapour density of gas is :-

(1) 32 (2) 40

(3) 16 (4) 8

74. Calculate the number of electrons present in 18 mL of water.

(1) 10 (2) 18

(3) 10 N_A (4) 18 N_A

75. For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, 28 g N_2 and 67.2L of H_2 at NTP are reacted. Calculate the volume of NH_3 formed at NTP.
(1) 22.4 L (2) 44.8 L
(3) 34 L (4) 89.6 L

76. A compound contains 20% oxygen. Then minimum molecular weight of compound will be :-
(1) 40 (2) 20
(3) 80 (4) 100

77. The equivalent weight of a metal is 4.5 and the molecular weight of its chloride is 80. The atomic weight of the metal is :-
(1) 18 (2) 9
(3) 4.5 (4) 36

78. 0.45g of acid (molecular weight = 90) was exactly neutralised by 20 mL of 0.5 N NaOH. Basicity of the acid is :-
(1) 2 (2) 1
(3) 4 (4) 3

79. When $KMnO_4$ reacts with KI in neutral medium then the product obtained is :-
(1) I_2 (2) IO_3^-
(3) IO_4^- (4) I^-

80. The highest stable oxidation state shown by actinoids is :-
(1) +3 (2) +4 (3) +5 (4) +6

81. For which of the following element, $E^\circ_{M^{+2}/M}$ value is positive ?
(1) Sc (2) Mn (3) Cu (4) Zn

82. Eu^{+3} can act as :-
(1) Oxidising agent
(2) Reducing agent
(3) Both
(4) None of these

83. Which of the following products are not obtained when lanthanoids are heated with carbon ?
(1) LnC_2 (2) Ln_2C_3
(3) Ln_3C_2 (4) Ln_3C

84. Which of the following match is incorrect ?
(1) Malachite green- $CuCO_3 . Cu(OH)_2$
(2) Calamine- $MgCO_3$
(3) Limestone- $CaCO_3$
(4) Carnallite- $KCl . MgCl_2 . 6H_2O$

85. Which reaction is correct with respect to Mc-Arthur forest process ?
(1) $3AgNO_3 + [Co(NH_3)_6]Cl_3 \rightarrow 3AgCl + [Co(NH_3)_6](NO_3)_3$
(2) Au + Aqua regia $\rightarrow AuCl_3$
(3) $Au + CN^- \rightarrow [Au(CN)_2]^-$
(4) None of these

86. Which of the following can act as strongest oxidising agent in aq. medium ?
(1) Cr^{+3} (2) Mn^{+3}
(3) Fe^{+3} (4) Co^{+3}

87. Which of the following is used to acidify $KMnO_4$?
(1) Conc. H_2SO_4
(2) dil. H_2SO_4
(3) dil. HCl
(4) All of these

88. During electrorefining of copper, which of the following elements are obtained as anode mud ?
(1) Na, K (2) Ag, Au
(3) Zn, Cu (4) Mg, Fe

89. Which of the following is not a method of concentration of ore ?
(1) Leaching
(2) Polling
(3) Gravity separation
(4) Magnetic separation

90. Maximum spin-only magnetic moment is shown by ?
(1) Eu^{+3} (2) La^{+3}
(3) Lu^{+3} (4) Gd^{+3}

TOPIC : Plant Morphology - Flower to Families, Plant Anatomy, Plant Diversity upto History of Taxonomy, Circulation, Excretion, Digestion

91. The mode of arrangement of sepals or petals in floral bud with respect to the other members of the same whorl is known as :-

(1) Placentation (2) Aestivation

(3) Adhesion (4) Cohesion

92. In morphology of flower which of the following statement is incorrect ?

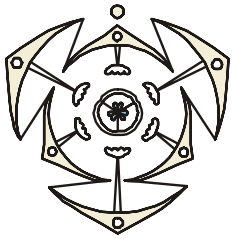
(1) In Pea plant two bundles of stamens formed

(2) A sterile stamen is called staminode.

(3) In salvia, filaments of stamens are of same length.

(4) Epitepalous condition is found in lily.

93. Identify the family with its character from the given below floral diagram :-



(1) Liliaceae, non-endospermic seed

(2) Liliaceae, bulb and corm

(3) Solanaceae, bulb and corm

(4) Fabaceae, Zygomorphic flower

94. Which of the following is correct for the given diagram below :-



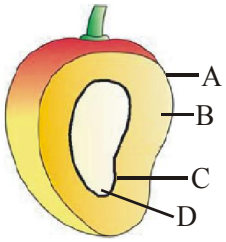
(1) Valvate - *Calotropis*

(2) Twisted - China rose

(3) Twisted - Pea

(4) Imbricate - Cotton

95. Given figure represent a drupe fruit select the option that correctly identifies A, B, C, D :-



	A	B	C	D
(1)	Pericarp	Epicarp	Mesocarp	Endocarp
(2)	Epicarp	Mesocarp	Endocarp	Seed
(3)	Mesocarp	Epicarp	Endocarp	Seed
(4)	Epicarp	Mesocarp	Seed	Endocarp

96. Study carefully the given floral diagram and select the option which correctly represents the related floral formula :-



(1) % ♀ $K_{(5)}\ C_{1+2+2}\ A_{9+1}\ \underline{G}_1$

(2) ⊕ ♀ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$

(3) % ♀ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_1$

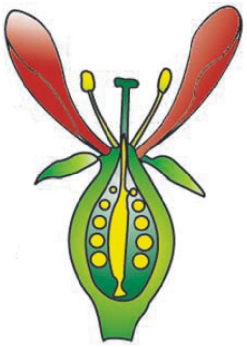
(4) ⊕ ♀ $K_5\ C_{1+(2)+2}\ A_{9+1}\ \underline{G}_1$

97. Pick up the correct match :-

Fruit		Family	
(I)	Cypsella	(A)	Poaceae
(II)	Caryopsis	(B)	Compositae
(III)	Pepo	(C)	Papilionatae
(IV)	Legume	(D)	Cucurbitaceae

	A	B	C	D
(1)	II	I	III	IV
(2)	IV	III	I	II
(3)	I	II	IV	III
(4)	II	I	IV	III

98. Carefully observe the given diagram and choose **incorrect** statement regarding this diagram.



(1) Thalamus grows upward completely and getting fused with it
(2) Ovary is said to be superior
(3) Ovary is said to be inferior
(4) It is found in guava and cucumber

99.



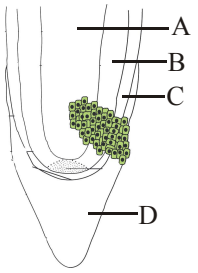
In this type of arrangement two smallest anterior petals are known as_A_ and found in_B_. A and B are respectively :-
(1) Vexillum and Pea (2) Keel and Tulip
(3) Wings and Lupin (4) Keel and Lupin

100. A common structural feature of vessel elements and sieve tube elements is :-
(1) Presence of p-protein
(2) Thick secondary walls
(3) Enucleate condition
(4) Pores on lateral walls

101. Inter fascicular cambium and cork cambium are formed due to :-
(1) Differentiation (2) De-differentiation
(3) Re-differentiation (4) Elongation

102. Select the true statement :-
(1) Bast fibres are parenchymatous cells.
(2) Monocot roots undergo secondary growth.
(3) Bulliform cells are modified abaxial epidermal cells in grasses which are small, empty and coloured cells.
(4) Sapwood involved in conduction of water and minerals from root to leaf.

103. Select the option that correctly identifies the labellings A, B,C, D in the given figure showing section of root apical meristem.



	A	B	C	D
(1)	Central cylinder	Cortex	Protoderm	Root cap
(2)	Protoderm	Cortex	Central cylinder	Root cap
(3)	Hypodermis	Epidermis	Cortex	Tunica
(4)	Cortex	Protoderm	Central cylinder	Root cap

104. Phellogen cuts off derivatives on the inner side to form ....(A).... and on the outer side to form....(B).....:-
(1) A–Cork ; B–Secondary Cortex (2) A–Secondary Cortex ; B-Cork (3) A–Cork Cambium ; B–Cork (4) A–Cork Cambium ; B–Secondary cortex

105. Which plant part possesses diarch to hexarch condition of vascular bundles with less developed pith ?
(1) Monocot root (2) Monocot stem
(3) Dicot stem (4) Dicot root

106. As compared to springwood, autumn wood has:-
(1) More number of xylary elements with wider vessels
(2) More number of xylary elements with narrow vessels
(3) Fewer xylary elements with wider vessels
(4) Fewer xylary elements with narrow vessels

107. Which of the following statements is **correct** about a woody dicot stem which shows extensive secondary growth ?
(1) Primary xylem persists in the centre of the axis
(2) Primary and the older secondary phloem get crushed
(3) Secondary xylem forms the bulk of the stem
(4) All of these

108. Read the following statements :-
(i) Multicellular epidermal hair
(ii) Collenchymatous hypodermis
(iii) Pith present
(iv) Vascular bundles present in a ring
Above given features describe which of the following plant part ?
(1) Monocot stem (2) Monocot root
(3) Dicot stem (4) Dicot root

109. Select true statements for wood found in dicots:-
(i) Heteroxylous wood (ii) Vessels absent
(iii) Soft wood (iv) Porous wood
(1) (i), (ii) & (iii) (2) (i) & (iv)
(3) (ii) & (iii) (4) (ii), (iii) & (iv)

110. Identify the type of vascular bundles given in the figures (A) and (B) and select the **correct** option:-

(A)  (B) 

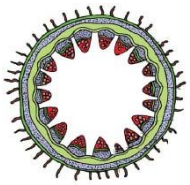
	(A)	(B)
(1)	conjoint, collateral	conjoint, bicollateral
(2)	conjoint, bicollateral	conjoint, collateral
(3)	conjoint, collateral, open	conjoint, collateral, closed
(4)	conjoint, collateral, closed	conjoint, collateral, open

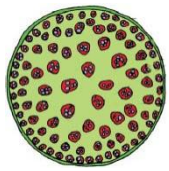
111. Which of the following statements are **incorrect** ?
(a) The two kinds of woods that appear as a alternate concentric rings constitute an annual ring
(b) The spring wood is darker in colour and has a higher density.
(c) In spring season, cambium is very active and produces a large number of xylary elements having vessels with wider cavities
(d) During rhytidome formation new phellogen develops outside the old one.
(1) b, c (2) a, c (3) b, d (4) b, c,d

112. In a dicot root, pericycle gives rise to:-
(1) Lateral roots
(2) Cork cambium
(3) A portion of vascular cambium
(4) All of the above

113. In dorsiventral leaf, location of palisade tissue and protophloem is respectively on _____surfaces :-
(1) Adaxial, Abaxial (2) Adaxial, Adaxial
(3) Abaxial, Abaxial (4) Abaxial, Adaxial

114. Indentify the given transverse sections of plant part and suggest the nature of xylem and order of development of xylem in vascular bundles of the respective sections :-


(A)


(B)

	Transverse section-A			Transverse section-B		
		Nature of xylem	Order of Development		Nature of xylem	Order of Development
(1)	Monocot Root	Exarch	Centripetal	Dicot Root	Exarch	Centripetal
(2)	Dicot stem	Exarch	Centripetal	Monocot stem	Exarch	Centripetal
(3)	Dicot stem	Endarch	Centrifugal	Monocot stem	Endarch	Centrifugal
(4)	Monocot stem	Endarch	Centrifugal	Dicot stem	Endarch	Centrifugal

115. How many tissue are the result of re-differentiation in dicot root ?
Intrafascicular cambium, Interfascicular cambium, Primary medullary rays, Secondary medullary rays, Vascular cambium, Secondary xylem, Secondary phloem, Phellem, Phelloderm, Phellogen.
(1) Five (2) Six (3) Seven (4) Eight

116. Identify the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem :-

(i) Secondary cortex (ii) Autumn wood
(iii) Secondary phloem (iv) Primary phloem
(1) (ii), (iii), (iv), (i) (2) (i), (ii), (iv), (iii)
(3) (i), (iv), (iii), (ii) (4) (iv), (iii), (i), (ii)

117. A cut trunk shows 26 concentric rings of spring wood and autumn wood in alternate rows. The age of trunk would be :-

(1) 26 years (2) 52 years
(3) 13 years (4) 104 years

118. Find the **correct** statements from the followings

(A) Each rank or taxon, in fact represents a unit of classification
(B) Every step in taxonomic hierarchy represents a rank or category
(C) A taxonomic category is a part of ovarall taxonomic hierarchy
(D) Taxonomic categories are not distinct biological entities, they are only morphological aggregates
(1) A, B & D are correct
(2) A, C & D are correct
(3) A, B & C are correct
(4) A and B are correct

119. Match the following and choose the correct option:-

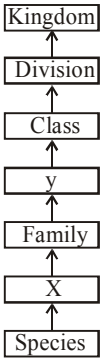
	Column-I		**Column-II**
(A)	Family	(i)	Dicotyledonae
(B)	Kingdom	(ii)	Polymoniales
(C)	Order	(iii)	Solanum
(D)	Class	(iv)	Plantae
(E)	Genus	(v)	Solanaceae

A	**B**	**C**	**D**	**E**
(1) (i)	(ii)	(iii)	(iv)	(v)
(2) (iv)	(v)	(ii)	(i)	(iii)
(3) (v)	(iv)	(ii)	(i)	(iii)
(4) (v)	(iii)	(ii)	(i)	(iv)

120. Select the mismatch from following :-

(1) Key → used for identification of plants and animals
(2) Monograph → contain information on any one taxon
(3) Museums → collections of preserved plant specimens only
(4) Herbarium → storehouse of collected plant specimens

121. Observe the gradation of taxonomic categories and identify the missing categories according to the statements given below :-



(i) x is a group of related species.
(ii) y is a group of related classes.
(iii) x is a group of related families.
(iv) y is a group of related divisions.
(v) y is a group of related families.
The **correct** options are :-
(1) i and iii (2) iii and ii
(3) ii, iv and v (4) i and v

122. Tiger, cat and dog are placed in same :-

(i) Family (ii) Class
(iii) Order (iv) Genus
(v) Species
(1) (i) & (iv) (2) (ii) & (iii)
(3) (i) & (iii) (4) (iv) & (v)

123. Which of the following names are not valid according to ICBN ?

(i) Solanum Tuberosum
(ii) *Mangifera indica* Linn
(iii) Mangifera mangifera
(iv) *Solanum tuberosum Linn*
(1) only (ii) (2) (ii) & (iv)
(3) (i), (iii) & (iv) (4) (i) & (iii)

124. Minimum amount of urea is present in :-

(1) Renal Artery (2) Hepatic Artery
(3) Renal vein (4) Hepatic vein

125. Jaundice is characterise by the rise of blood level of:-
(1) Bile salt (2) Bile pigments
(3) Bile juice (4) Stercobilin

126. Thin segment of Ascending limb of loop of Henle is the lined by ?
(1) Cuboidal cells (2) Squamous cells
(3) Columnar cells (4) None of these

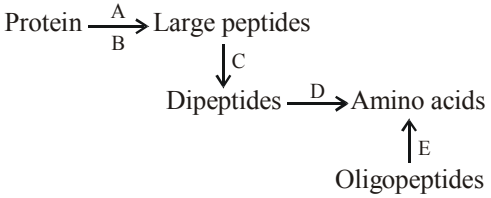
127. Read the following points :-
(i) Decreases GFR
(ii) Increases B.P.
(iii) Decreases blood volume
(iv) Increases aldosterone secretion
(v) Vasodilator
(vi) Increases $Na^{\oplus}$ excretion
How many points are correct about ANF activity ?
(1) Two (2) Three (3) Four (4) Five

128. Which one of the following is correct regarding the excretion ?
(1) Maximum amount of water from renal filtrate is reabsorbed in DCT.
(2) The descending limb of loop of Henle is completely impermeable to NaCl.
(3) Malpighian corpuscle is found in medulla region of kidney.
(4) The colour of Fresh urine is pale yellow and is slightly alkaline in nature.

129. Read the following five statement and answer as asked next to them ?
(a) Digestion of protein is accomplished in ileum
(b) Enterokinase is a nondigestive enzyme
(c) Haustra are enlarged rectal veins
(d) Zinc is required for carboxy peptidase activity.
(e) Spicy food may cause indigestion
How many of the above statements are correct?
(1) Four (2) One (3) Two (4) Three

130. How many enzymes in the list given below acts on peptide bond ?
Chymotrypsin, Aminopeptidase, Pepsin, Lactase, Rennin, Carboxypeptidase, Nuclease, Nucleotidase:-
(1) Six (2) Three (3) Five (4) Four

131. Identify true statement about enzymes A, B, C, D and E from the following :



Hints :-
(a) Enzyme A is activated in stomach
(b) Enzyme B is activated by trypsin
(c) Enzyme C is an enzyme of pancreatic juice
(d) Enzyme D and E are present in succus entericus
(1) A = Renin ; C = Carboxypeptidase
(2) A = Rennin ; C = Carboxypeptidase
(3) B = Trypsinogen ; E = Steapsin
(4) B = Amylase ; C = Carboxypeptidase

132. The typical Lubb-Dup sounds heard in the heart beat of a healthy person are due to :-
(1) Closing of the tricuspid and bicuspid valve
(2) Blood flow through the aorta
(3) Closing of the tricuspid and semilunar valves
(4) Closing of the semilunar valves

133. The Rh antibodies from the mother can leak into the blood of fetus and destroy the foetal RBCs. This condition is called :-
(1) Erythroblastosis foetalis
(2) Anaemia
(3) Erythropenia
(4) Leukemia

134. Complete double circulation is found in :-
(1) Fish, birds, mammals
(2) Crocodile, reptiles, amphibians
(3) Crocodile, birds, mammals
(4) Amphibians, crocodile, mammals

135. In a cardiac cycle, time period of atrial diastole is:-
(1) 0.1 sec. (2) 0.3 sec.
(3) 0.5 sec. (4) 0.7 sec.

136. When do AV-valves closes during the cardiac cycle?
(1) when the pressure inside the ventricles is at the lowest
(2) when the pressure inside the atria is greater than that inside the ventricles
(3) when ventricular pressure exceeds that of the atria
(4) when the pressure is greater in the aorta than in the left ventricle

137. Glucose and urea are present in :-
(1) Only in blood
(2) Blood and urine
(3) Only in urine
(4) Blood and glomerular filtrate

138. Which is not reabsorbed actively in nephron?
(1) Glucose
(2) Na^{+}
(3) Amino acid
(4) Nitrogenous wastes

139. Angiotensin–II activates :-
(1) Atria of heart to release ANF
(2) Distal part of tubule to reabsorb sodium and glucose.
(3) Adrenal cortex to release aldosterone.
(4) JG-Cells to secrete renin

140. Diabetes insipidus is not characterized by :
(1) Dehydration (2) Polyuria
(3) Polydipsia (4) Glycosuria

141. In ornithine cycle, which of the following wastes are removed from the blood –
(1) CO_2 and ammonia
(2) Ammonia and urea
(3) CO_2 and urea
(4) Urea and urine

142. Given below a table in which name of digestive enzymes, their substrate and their products are given in column I, II and III respectively. Select the option in which correctly matched pairs are given ?

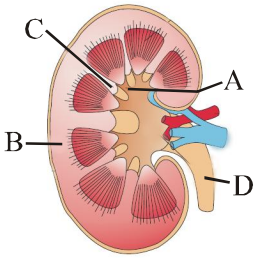
S. No.	Column-I Name of enzyme	Column-II Substrate	Column-III Product
(i)	Maltase	Maltose	Glucose + Fructose
(ii)	Lactase	Lactose	Glucose + galactose
(iii)	Nucleosidase	Nucleotide	Sugar +N- Bases
(iv)	Sucrase	Sucrose	Glucose + Fructose
(v)	Nucleotidase	Nucleotide	Nucleoside + phosphate group

Options :-
(1) (i), (ii), (iv) & (v)
(2) (i), (ii), (iii) & (v)
(3) (ii), (iv) & (v)
(4) (i), (iii) & (iv)

143. Which part of nephron is impermeable to water:-
(1) Descending limb of Henle's loop
(2) Ascending limb of Henle's loop
(3) PCT
(4) DCT

144. Bundle of His is a network of :-
(1) Muscle fibres distributed throughout the heart walls
(2) Muscle fibres found only in the ventricle wall
(3) Nerve fibres distributed in ventricles
(4) Nerve fibres found throughout the heart

145. Given is the figure of longitudinal section of kidney. Identity the parts labelled as A to D and select the **correct** option :-



	A	B	C	D
(1)	Cortex	Calyx	Renal column	Ureter
(2)	Calyx	Cortex	Renal column	Ureter
(3)	Medulla	Cortex	Renal capsule	Urethra
(4)	Calyx	Cortex	Renal pelvis	Urethra

146. Which is correct with respect to all leucocytes :-
(1) Neutrophils - 2-3 lobed nucleus, 20% - 25%
(2) Monocytes - bilobed nucleus, 5% - 8%
(3) Basophils - 2-3 lobed nucleus, 0.5% - 1%
(4) Acidophils - 3-7 lobed nucleus, 2% - 3%

147. Which of the following is correct option for Blood groups :-

	Blood Group	Antigens on RBCs	Antibodies in plasma
(1)	A	A	Anti-B
(2)	B	A	Anti-A
(3)	AB	Nil	Anti-A,B
(4)	O	A,B	Nil

148. How many statement are incorrect for WBC ?
(A) They are colourless due to lack of Haemoglobin
(B) They are are non-nucleated in most of the mammals
(C) Avg count is 6000-8000 mm^3 of blood
(D) Generally long life span in blood
(1) 2 (2) 3 (3) 1 (4) 4

149. If following data belongs to an adult male

$$\text{Cardiac index} = \frac{\text{Cardiac output}}{\text{Total body surface area}}$$

Total body surface area in adult = $1.7m^2$
Cardiac index – 4 litre/mt./m^2
Heart rate – 68/minute
End systolic volume – 50 ml.
Then choose the correct set of option.
(1) Cardiac output = 5 litre/minute., stroke volume = 70 ml. ; EDV = 120 ml.
(2) C.O. = 5 litre/minute. ; SV = 100 ml.; EDV = 50 ml.
(3) C.O. = 6.8 litre/minute., SV = 50 ml; EDV = 120 ml.
(4) C.O. = 6.8 litre/minute. ; SV = 100ml.; EDV = 150 ml.

150. Match column–I with column–II :-

	Column–I		Column–II
(A)	Cardiac arrest	(i)	Heart not pumping blood effectively
(B)	Heart failure	(ii)	Heart muscle is suddenly damaged
(C)	Heart attack	(iii)	Acute chest pain
(D)	Angina	(iv)	Heart stops beating

(1) (A)–i, (B)–ii, (C)–iii, (D)–iv
(2) (A)–iv, (B)–ii, (C)–i, (D)–iii
(3) (A)–iv, (B)–i, (C)–ii, (D)–iii
(4) (A)–ii, (B)–iii, (C)–i, (D)–iv

151. Which are not correctly matched :-
(A) Normal blood pressure = 80/120 mm Hg
(B) Stroke volume = 70 ml
(C) Cardiac output = 5000 ml
(D) Breathing rate = 44-46 times/min.
(E) Heart beat rate = 72 times/min.
(F) Duration of a cardiac cycle = 0.8 sec
(1) A, B, D, F (2) B, C, E
(3) A, D (4) Only D

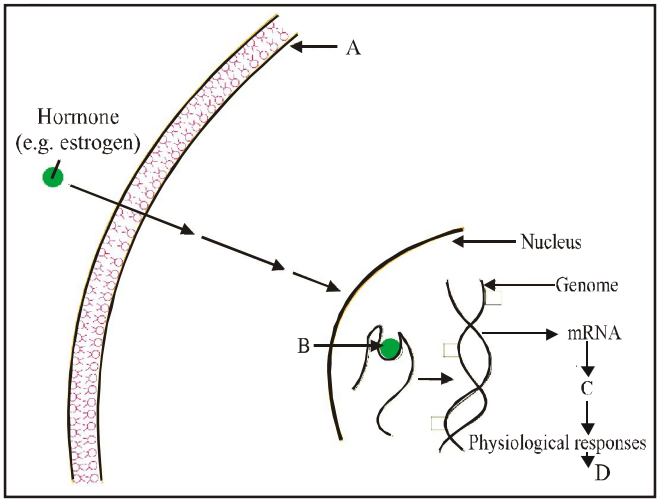
152. Read the following four statements (A-D) :-
(A) In amphibia and reptiles incomplete double circulation present.
(B) In arthropods and molluscs open circulatory system present.
(C) In joint diastole stage semilunar valves are closed.
(D) The hepatic portal vein carries blood from intestine to the kidney.
How many of the above statements are wrong?
(1) Four (2) One (3) Two (4) Three

153. Which of the following statement are False/True?
A. PTH regulates the metabolism of Ca^{2+} and PO_4^-
B. Prolactin inhibits the action of GnRH
C. Dopamine inhibits lactation
D. Insulin promotes lipolysis
(1) A and B are true, C and D are false
(2) A and C are true, B and D are false
(3) A, B and C are true, D only false
(4) B, C and D are false, A only true.

154. How many hormone in given list are related with immune system ?
Thyroxine, Aldosterone, Thymosine, Glucagon, Growth hormone, Cortisole, Testosteron.
(1) Six (2) Four (3) Two (4) One

155. Grave's disease occur due to :-
(1) Hypo secretion of hypothalamic hormone
(2) Hypo secretion of corticoides hormone
(3) Hypo secretion of pituitary hormone
(4) Hyper secretion of thyroid hormone

156. Identify the structure labelled as A,B,C and D in the following figure, showing the mode of action of a steroid hormone. :-



(1) A-Uterine cell membrane, B-Receptor protein, C-CAMP
(2) A-Ovarian cell membrane, B-Hormone-receptor complex, D-Biochemical response
(3) B-C-AMP, C-Protein, D-Tissue growth and differentiation
(4) A-Uterine cell membrane, B-Hormone-receptor complex, D-Tissue growth and differentiation

157. Louis Pasteur's experiment disproves the:
(1) Big-bang theory
(2) Theory of biogenesis
(3) Theory of abiogenesis
(4) Oparin-Haldane theory

158. According to modern and most accepted theory of origin of life, life came from:
(1) Outer space
(2) Dead and decaying matter
(3) Non-living organic macromolecules
(4) Created by god

159. The correct combination of gases and temperature taken by S. Miller for his experiment was:
(1) Methane, Carbon-di-oxide and Ammonia at 800ºC
(2) Methane, Ammonia and Hydrogen sulphide at 5000ºC
(3) Hydrogen, Ammonia and water vapour at 800ºC
(4) Methane, Ammonia, Hydrogen and water vapour at 800ºC

160. Which of the following is an example of convergent evolution?
(1) Members of a same population showing diversity in their food
(2) Members of two different species adapted for different habitats
(3) Members of same species adapted for same habitats
(4) Members of different species adapted for same habitats

161. Baer provided what kind of evidences in favour of organic evolution?
(1) Embryological evidences
(2) Biogeographical evidences
(3) Palaentological evidences
(4) All of the above

162. Which of the following option represents analogous structures?
(1) Hands of human and flippers of whale
(2) Thorn of Bougainvillea and tendril of Cucurbita
(3) Edible parts of radish and carrot
(4) Gills of fishes and lungs of birds

163. The first organisms that invaded land, were:
(1) Coelacanth (2) Amphibians
(3) Reptiles (4) Plants

164. During evolution some land reptiles went back into water to evolve into fish like reptiles. These animals are known as:
(1) Ichthyosaurs (2) Lobefins
(3) Blue whales (4) Crocodiles

165. Those who are better fit in an environment:
(1) leave more progeny than others
(2) will survive more
(3) are selected by nature
(4) all of the above

166. Evolution by natural selection, in a true sense started when:
(1) Universe originated
(2) Earth originated
(3) Human originated
(4) Cellular forms of life originated

167. When natural selection leads to stabilization, than in frequency distribution curve of a population:
(1) more individuals acquire mean character value
(2) more individuals acquire value other than mean character value
(3) more individuals acquire peripheral character value at both ends of the curve
(4) individuals acquire all character values equally

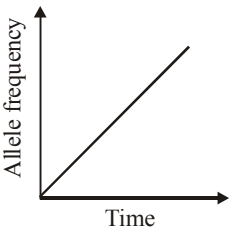
168. Which of the following statement is correct about *Homo erectus*?
(1) Its fossils were discovered from Java
(2) It lived in East-African grasslands
(3) It was first human-like being
(4) It started agriculture

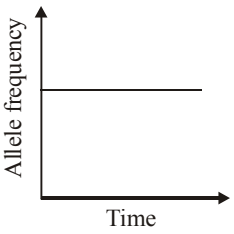
169. The by chance change in allelic frequency of a small, isolated population is an example of:
(1) Gene migration
(2) Gene flow
(3) Gene recombination
(4) Genetic drift

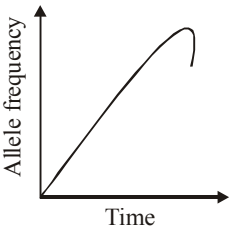
170. What isn't true about evolution?
(1) It is a stochastic process
(2) It depends on chance events in nature
(3) It depends on chance mutations in organisms
(4) It is a directed process in the sense of determinism

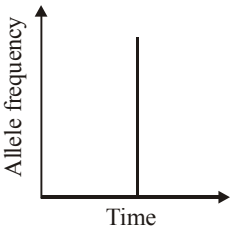
171. Which is the most common mechanism of genetic variations in the population of sexually reproducing organism?
(1) Point mutation
(2) Chromosomal aberrations
(3) Transition
(4) Gene recombination

172. Which graph represents a population in genetic equilibrium according to Hardy-Weinberg principle?

(1) 

(2) 

(3) 

(4) 

173. Viceroy butterfly escapes from predators by:

(1) Foul smell (2) Bitter taste

(3) Mimicry (4) Rough skin

174. Which is not correctly matched?

(1) Natural selection – Charles Darwin

(2) Use and Disuse of organs – Lamarck

(3) Theory of Biogenesis – Hugo deVries

(4) Swan neck flask experiment – Louis Pasteur

175. The milk yield in a dairy farm is primarily dependent on:

(1) Quantity of fodder

(2) Quality of fodder

(3) Cleanliness of farm

(4) Quality of breed

176. Which is an exotic breed of high milk yielding breed of cow?

(1) Merino

(2) Jersey

(3) Bikaneri

(4) Leghorn

177. If we want to evolve a pureline in animals, which type of mating should be carried out?

(1) Inbreeding

(2) Out-crossing

(3) Alternative inbreeding and out-breeding

(4) Cross-breeding

178. Mules are the result of:

(1) Inbreeding

(2) Out-crossing

(3) Cross-breeding

(4) Interspecific hybridization

179. Select the correct option from column I and column II about the different forms of honeybee:

Column-I	Column-II
(a) Queen	(i) Haploid
(b) Drone	(ii) Egg laying
(c) Workers	(iii) Sterile
	(iv) Secretion of Royal jelly
	(v) Largest in size
	(vi) Maximum in numbers
	(vii) Sting absent
	(viii) Minimum in numbers
	(ix) Males
	(x) Nourishment of larvae

Options :

(1) a-ii,iv,v,vii,viii; b-i,iii,ix; c-vi,x

(2) a-ii,v,viii; b-i,vii,ix; c-iii,iv,vi,x

(3) a-iii,v,vii; b-i,viii,ix; c-ii,iv,vi,x

(4) a-ii,vi,vii; b-i,viii,x; c-iii,iv,v,ix

180. Which is a fresh water fish among the following?

(1) Hilsa (2) Sardines

(3) Mackerel (4) Catla

SPACE FOR ROUGH WORK

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS

(Session : 2019 - 2020)

ENTHUSIAST COURSE : PHASE : MEQ

MINOR TEST NO.	CURRENT UNIT	PREVIOUS SYLLABUS			
		PREVIOUS UNIT	FROM TOPIC		
09 29-12-2019	Syllabus Covered During 02-12-2019 to 28-12-2019	Syllabus Covered (Current Portion) in Minor Test Dated 01-12-2019	P	MODERN PHYSICS	SUMIT VIJAY
				EMI, AC, EMW	ATUL SHUKLA
			C	CHEMICAL KINETICS, SOLUTION, SURAFCE CHEMISTRY	DEVPARSHVA ZABAKH
			B	LIFE CYCLE OF ANGIOSPERM, PLANT ANATOMY, MORPHOLOGY	S. D. SHARMA

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

ENTHUSIAST COURSE (PHASE : MEQ)

01–12–2019

PATTERN / TYPE : NEET / MINOR — ANSWER KEY — PRE-MEDICAL 2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	2	2	4	4	2	4	2	3	3	2	2	1	4	1	3	3	3	3	2	3	4	2	3	4	3	3	3	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	3	4	4	3	3	2	2	2	3	2	2	1	2	2	2	4	2	4	1	2	2	2	3	3	4	2	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	3	1	1	1	2	2	1	4	4	4	3	3	2	3	2	1	2	4	3	1	3	2	3	4	2	2	2	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	2	2	2	3	4	2	4	3	2	4	1	2	4	4	4	3	2	4	3	4	1	3	2	3	3	3	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	3	3	2	2	3	2	4	3	2	3	1	3	4	3	4	4	3	4	1	3	2	2	2	3	1	1	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	3	3	4	4	3	3	4	4	1	4	4	1	4	4	1	1	4	4	4	2	3	3	4	2	1	4	2	4

ENTHUSIAST COURSE

PHASE : MEQ

TARGET : PRE-MEDICAL 2020

Test Type : **MINOR** Test Pattern : NEET(UG)

TEST DATE : 01 - 12 - 2019

HINT – SHEET

1. $v = \frac{dx}{dt} = 32 - 8t^2$

$v = 0$ at $t = 2s$

$a = \frac{dv}{dt} = -16t$

at $t = 2s$, $a = -32\ m/s^2$

2. $h = \frac{1}{2}gT^2$

At $\frac{T}{3}$ second, distance fallen

$d = \frac{1}{2}g\left(\frac{T}{3}\right) = \frac{h}{9}$

$\therefore$ Height from ground $= h - d = \frac{8h}{9}$

3. **At 4 s**

$u = at = 8m/s$

$s_1 = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 4^2 = 16\ m$

From 4 s to 8 s

$a = 0$, $v =$ constant $= 8\ m/s$

$\therefore \quad s_2 = vt = (8)(4) = 32\ m$

From 8 s to 12 s

$s_3 = s_1 = 16\ m$

$\therefore \quad s_{Total} = s_1 + s_2 + s_3 = 64\ m$

4. $v^2 = 25 + 25s$

or $v^2 = (5)^2 + 2\ (12.5)\ s$

Now compare with $v^2 = u^2 + 2as$

5. $T_1 = \frac{2u\sin\theta}{g}$, $T_2 = \frac{2u\cos\theta}{g}$

$R = \frac{2(u\sin\theta)(u\cos\theta)}{g}$

$= \frac{2\left(\frac{gT_1}{2}\right)\left(\frac{gT_2}{2}\right)}{g} = \frac{1}{2}gT_1T_2$

6. $H_1 = \frac{u^2\sin^2\theta}{2g}$

$\therefore \quad 102 = \frac{(u^2)\sin^2 60°}{20}$

$\therefore \quad u = 52.2\ m/s$

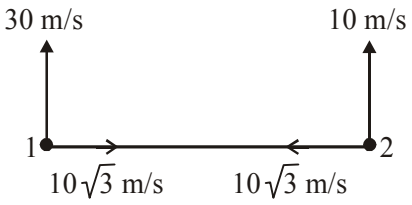
Other stone should be projected at $90° - \theta$ or $30°$ from horizontal.

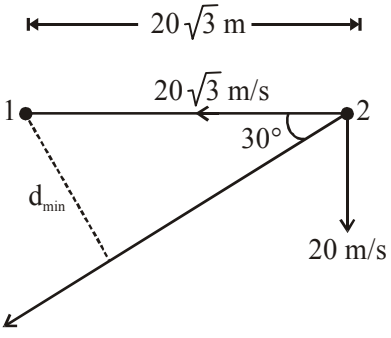
$\therefore \quad H_2 = \frac{u^2\sin^2 30°}{2g}$

$= \frac{(52.2)^2(1/4)}{20}$

$= 34\ m$

7. $a_1 = a_2 = g$ (downwards)





$\therefore \quad a_{12} = 0$

$\therefore$ Relative motion between them is uniform.

Relative velocity v_{21}

$d_{min} = 20\sqrt{3}\ \sin 30°$

$= 10\sqrt{3}\ m$

8. $A = A_0e^{-\lambda t}$

$t_1 = t$

$t_2 = 2T$

$$\frac{A_0}{3} = A_0e^{-\lambda t}$$

$$e^{-\lambda T} = \frac{1}{3} \quad ...(1)$$

$A = A_0e^{-\lambda(2T)}$

$A = A_0(e^{-\lambda T})^2$

$$A = \frac{A_0}{9}$$

9. $$KE = \frac{3}{4}TE$$

$$\frac{1}{2}K\left(A^2 - x^2\right) = \frac{3}{4}\frac{1}{2}KA^2$$

$$A^2 - x^2 = \frac{3}{4}A^2$$

$$x^2 = \frac{A^2}{4}$$

$$\boxed{x = \frac{A}{2}}$$

for $\frac{A}{2}$, $\phi = \frac{\pi}{6}$

$$t = \frac{T}{12} = \frac{2}{12} = \frac{1}{6}s$$

10. $$\frac{T_1}{T_2} = \sqrt{\frac{M}{m+M}}$$

$$\frac{T}{\frac{5T}{3}} = \sqrt{\frac{m}{m+M}}$$

$$\frac{9}{25} = \sqrt{\frac{m}{m+M}}$$

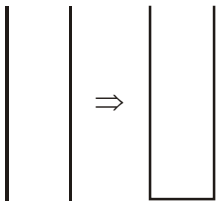
$9m + 9M = 25M$

$9m = 16m$

$$\frac{m}{M} = \frac{16}{9}$$

11. $$\Delta n = \frac{2nv_s}{v}$$

14. Now given that



f $\quad$ f /2 → Fundamental frequencies

$$\frac{3f}{2} - f = 100 \text{ or } \frac{f}{2} = 100$$

$\therefore$ f = 200 Hz

17. $$A_R^2 = 10^2 + 20^2 + 2.10.20.\cos\frac{\pi}{2}$$

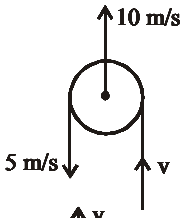
$$A_R^2 = 500$$

$$A_R = 10\sqrt{5}$$

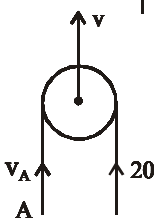
18. $$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{4}{1}$$

19. With 514 Hz, two beats are heard. Therefore frequency of tuning fork can be either 512 Hz or 516 Hz. With 510 Hz it gives beats of frequency 6 Hz. Hence, frequency of tuning fork is 516 Hz.

20.



$$10 = \frac{v-5}{2} \Rightarrow v = 25 \text{ m/s}$$



$$v = \frac{20 + v_A}{2} = 25 = \frac{20 + v_A}{2}$$

$\Rightarrow v_A = 30$ m/s

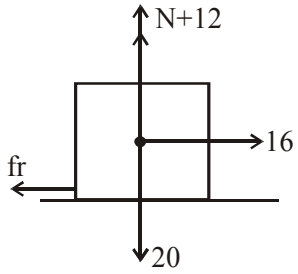
21. Area of F vs t graph = ΔP

$2(v_5 - v_0) = 12 + 4 = 16$

$v_5 - v_0 = 8$

$v_5 = 8 + v_0 = 13$ m/s

22.



N + 12 = 20

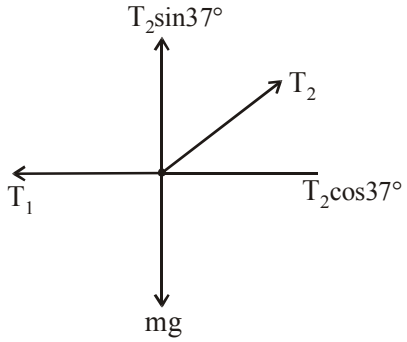
N = 8 Newton

fr = 1.6 N

$$a_B = \frac{16-1.6}{2} = \frac{14.4}{2} = 7.2 \text{ m/s}^2$$

$v_B = 0 + 5 \times 7.2 = 36$ m/s

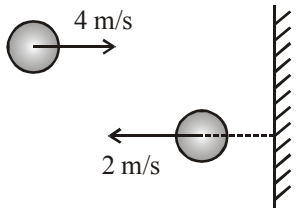
23.



$T_1 = T_2 \cos 37°$

$$\frac{T_1}{T_2} = \frac{4}{5}$$

24.



ΔP = impulse

I = 2(4 –(–2))

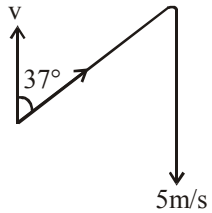
I = 12 N.s

25. All forces can provide impulse.

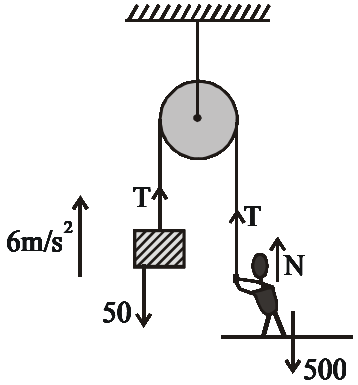
26. v cos37° = 5

$$v \times \frac{4}{5} = 5$$

$$v = \frac{25}{4} \text{ m/s}$$



27.



T – 50 = 5 × 6

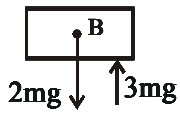
T = 80N

T + N = 500

N = 500 – 80

N = 420

28.



Net force by earth on block B is mg upward. So net force by B on earth is mg downward.

29. From work-energy theorem :

$\Delta KE = W_{ext}$

$$\Rightarrow \frac{1}{2} m[v^2 - u^2] = \int P dt$$

$$\Rightarrow \frac{1}{2} mv^2 = \int_0^2 \frac{3}{2} t^2 \cdot dt$$

$\Rightarrow v = 2$m/s

30. Applying work – energy theorem on the block, we get

$$F\ell - \frac{1}{2} k\ell^2 = 0$$

$$\Rightarrow \ell = \frac{2F}{K}$$

$$\therefore \quad \text{work done by } F = F\ell = \frac{2F^2}{K}$$

31. Force = $v\frac{dm}{dt} = v\left[\frac{d}{dt}(\text{volume} \times \text{density})\right]$

$$= v\frac{d}{dt}[Ax \times \rho] = \rho Av^2$$

$\therefore$ Power = Force × velocity

$= \rho Av^2 \times v = \rho Av^3$

$\therefore \quad \rho \propto v^3$

32. For conservative forces,

$$\Delta U = -\int_0^x F dx = -\int_0^x kx\, dx$$

$$U(x) - U(0) = -\frac{1}{2} kx^2$$

But $U(0) = 0 \Rightarrow U(x) = -\frac{1}{2} kx^2$ which represents a parabola passing through origin and mouth opening downwards

34. $V_x = -\frac{GM_1}{x} - \frac{GM_2}{R_2} = -G\left(\frac{M_1}{x} + \frac{M_2}{R_2}\right)$

35. $mg_d = mg\left(1 - \frac{d}{R}\right) = 0$

36. Change in gravitational PE $= -\frac{GMm}{4R} - \left(-\frac{GMm}{R}\right)$

$= \frac{3}{4}\frac{GMm}{R} = \frac{3}{4}mgR$

37. $v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{R+h}} = \frac{V_e}{2} = \frac{1}{2}\sqrt{\frac{2GM}{R}}$

$R + h = 2R \Rightarrow h = R$

38. By using COME

$-\frac{GMm}{R} + \frac{1}{2}mv_e^2 = -\frac{GMm}{4R} + \frac{1}{2}mv^2$

$\Rightarrow$ v = 5.6 km/s

39. Form the problem it is self-evident that the orbit of the comet is elliptic with sun being at one focus. Now as for elliptic orbits, according to Kepler's third law,

$$T^2 = \frac{4\pi^2}{GM}a^3, \text{ i.e., } a = \left(\frac{T^2GM}{4\pi^2}\right)^{1/3}$$

$$\text{or } a = \left[\frac{(76\times3.15\times10^7)^2\times6.67\times10^{-11}\times2\times10^{30}}{4\pi^2}\right]^{1/3}$$

$\simeq 2.7 \times 10^{12}$ m

But in elliptical orbits

$2a = r_{max} + r_{min}$

$r_{max} = 2a - r_{min}$

$= 2 \times 2.7 \times 10^{12} - 8.9 \times 10^{10}$

$= 5.3 \times 10^{12}$ m

40. $\vec{\tau} = \vec{M}\times\vec{B} = \left(50\hat{i}\right)\times\left(0.5\hat{i} + 3\hat{j}\right)$

$= 150\ \hat{k}$ N – m

42. $\frac{\mu_0 NI}{2R} = B_H \tan 60°$

$$I = \frac{2\times10^{-1}\times4\times10^{-5}\left(\sqrt{3}\right)}{4\pi\times10^{-7}\times10} = \frac{2\times1.732}{\pi} = 1.1A$$

43. As $r = \frac{\sqrt{2mK}}{qB}$

$\frac{\sqrt{2mK_P}}{eB} = \frac{\sqrt{2(4)mK_\alpha}}{2eB}$

$4[2mK_P] = 8mK_\alpha$

$K_\alpha = K_P = 1$MeV

91. NCERT–XI Pg. # 74, Para-5.5.1.2

92. NCERT–XI Pg. # 75, Para-5.5.1.3

93. NCERT–XI, Pg. # 81, Fig.-5.23(d)

94. NCERT–XI, Pg. # 74, Para-5.5.1.2, Fig.-5.15(b)

95. NCERT–XI, Pg. # 76, Fig.-5.17(a)

96. NCERT–XI, Pg. # 79, Fig.-5.21(f) Para-5.9.1

98. NCERT–XI, Pg. # 73, Para-5.5, Fig.-5.13(d)

99. NCERT–XI, Fig.-5.15(d)

100. NCERT–XI, Pg. # 87, 88 Para-6.1.2.2

102. NCERT–XI, Pg. # 88, 91, 94, 96, Para-6.1.2.2, 6.3.2, 6.3.6, 6.4.1.4

103. NCERT–XI, Pg. # 85, Fig. 6.1(a)

104. NCERT–XI, Pg. # 96, Para-6.4.2

106. NCERT–XI, Pg. # 96, Para-6.4.1.3

107. NCERT–XI, Pg. # 95, Para-6.4.1.2

110. NCERT–XI, Pg. # 90, Para-6.5 (b, c)

112. NCERT–XI, Pg. # 91, Para-6.3.1

114. NCERT–XI, Pg. # 92, Fig. 6.7 (a), (b)

118. NCERT–XI, Pg. # 8, Para-1.3

119. NCERT–XI, Pg. # 9, 10, Para-1.3.2, 1.3.3, 1.3.4, 1.3.5

120. NCERT–XI, Pg. # 11, 12, 13, 14, Para-1.4.1, 1.4.3, 1.4.5

121. NCERT–XI, Pg. # 10, Fig. 1.1

122. NCERT–XI, Pg. # 10, Para-1.3.4, 1.3.5

123. NCERT–XI, Pg. # 7, Para-1.2

157. NCERT(XII) Pg # 127/137(H) Para:7.1

158. NCERT(XII) Pg # 127/137(H) Para:7.1

159. NCERT(XII) Pg # 127/138(H) Para:7.1

161. NCERT(XII) Pg # 129/141(H) Para:7.3

163. NCERT(XII) Pg # 138/148(H) Para:7.8

164. NCERT (XII) Pg # 140/149(H) Para 7.8

165. NCERT(XII) Pg # 128/139(H) Para:7.2

166. NCERT(XII) Pg # 134/144(H) Para:7.5

167. NCERT(XII) Pg # 137/148(H) Para:7.7

168. NCERT(XII) Pg # 140/151(H) Para:7.9

169. NCERT(XII) Pg # 137/148(H) Para:7.7

170. NCERT(XII) Pg # 132/143(H) Para:7.3

175. NCERT(XII) Pg # 166/180(H) Para:9.1.1.1.

176. NCERT(XII) Pg # 167/181(H) Fig:9.1 (a)

177. NCERT(XII) Pg # 167/182(H) Para:9.1.2

178. NCERT(XII) Pg # 168/183(H) Para:9.1.2

180. NCERT(XII) Pg # 169/184(H) Para:9.1.4